

Foundation (Jalapeno)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** If two lines have slopes m_1 and m_2 , what is the condition for them to be perpendicular?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 + m_2 = 0$
(E) $m_1 - m_2 = 1$
- Q3.** Find the equation of a line perpendicular to $y = x - 2$ passing through $(1, 1)$.
(A) $y = -x + 2$
(B) $y = x + 1$
(C) $y = -x$
(D) $y = x - 1$
(E) $y = -x + 1$
- Q4.** Which of the following lines is perpendicular to $y = 3x + 2$?
(A) $y = \frac{1}{3}x - 1$
(B) $y = -\frac{1}{3}x + 1$
(C) $y = 3x - 2$
(D) $y = -3x + 1$
(E) $y = \frac{1}{3}x + 2$
- Q5.** What is the relationship between the slopes of two perpendicular lines?
(A) Equal
(B) Opposite
(C) Negative Reciprocal
(D) Parallel
(E) None
- Q6.** Find the slope of a line perpendicular to $y = -2x + 1$.
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) $\frac{1}{2}$
- Q7.** Which of the following is the equation of a line perpendicular to $y = x + 1$?
(A) $y = x - 1$
(B) $y = -x + 1$
(C) $y = x + 2$
(D) $y = -x$
(E) $y = -x - 1$
- Q8.** If m_1 and m_2 are the slopes of two perpendicular lines, then $m_1 \cdot m_2 =$
(A) -1
(B) 0
(C) 1
(D) 2
(E) -2

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line perpendicular to $y = 2x - 1$ and passing through $(2, 3)$. Show all work and explain your reasoning.
- Q10.** Prove that the slopes of two perpendicular lines are negative reciprocals of each other. Use the graph of $y = x$ as an example.

Chef's Tips

- Tip 1: Use the negative reciprocal slope property to identify perpendicular lines.
- Tip 2: When writing equations of perpendicular lines, use the point-slope form $y - y_1 = m(x - x_1)$.
- Tip 3: For free-response questions, make sure to show all work and explain your reasoning clearly.